

Current Internet, Networking and Cloud IaaS — Detailed Lecture Notes

JKSSB Computer Awareness

October 8, 2026

62.1 TCP: ordered and reliable, but with no timing promise

A running Internet conversation must survive lost, duplicated, and reordered packets. **TCP** (Transmission Control Protocol) is the transport protocol that hides this mess from the application. It is *connection-oriented*: the two ends set up a connection before any data moves. Once the connection exists, TCP numbers every byte, acknowledges what arrives, and retransmits whatever is lost, so the receiving application sees the bytes in exactly the order they were sent. That is the double guarantee to memorise: **ordered** delivery and **reliable** delivery.

What TCP does not promise is timing. It never claims the data will arrive within a fixed time, because retransmissions take as long as they take. This “no timing guarantee” point is tested directly (PYQ-18): an option that claims TCP promises delivery by a deadline is wrong.

Definition (TCP). *A connection-oriented transport protocol that delivers bytes in order and retransmits lost segments, with no guarantee about delivery time.*

Term	Meaning in this lesson
TCP	Transmission Control Protocol; ordered and reliable transport.
UDP	User Datagram Protocol; connectionless, fast, unreliable transport.
ICMP	Internet Control Message Protocol; control and diagnostic messages only.
BGP	Border Gateway Protocol; inter-domain routing between autonomous systems.
IPv6	128-bit next-generation Internet addressing; no broadcast.
QUIC	Quick UDP Internet Connections; the UDP-based transport under HTTP/3.
DNS	Domain Name System; translates domain names into IP addresses.
PIR	Public Interest Registry; operates the .org registry.
ICANN	Internet Corporation for Assigned Names and Numbers; global name/number coordination.
ARPANET	Advanced Research Projects Agency Network (1969); ancestor of the Internet.
IaaS	Infrastructure as a Service; provider manages hardware, customer manages OS and above.

The rest of this lesson uses these terms in one consistent sense. **TCP** means ordered and reliable transport. **UDP** means the fast, connectionless alternative with neither guarantee. **ICMP** means the control-message protocol that never carries application data. **BGP** means routing between autonomous systems, never inside one LAN. **IPv6** means 128-bit addresses with no broadcast. **QUIC** means the UDP-based transport that HTTP/3 uses instead of TCP. **DNS** means the name-to-address directory. **PIR** runs the .org registry under **ICANN** coordination. **ARPANET** is the 1969 packet-switched ancestor of the Internet. **IaaS** splits responsibility so that the provider manages networking hardware while the customer manages the operating system, applications, and data.

62.2 UDP and the TCP/UDP contrast

UDP (User Datagram Protocol) is TCP's near-neighbour and the most common distractor for it. UDP is *connectionless*: each datagram is sent independently with no setup. It provides no ordering guarantee, no reliability guarantee, and no retransmission. Its header is small, so it is faster and cheaper per packet than TCP.

Definition (UDP). *A connectionless transport protocol that sends independent datagrams with no ordering, reliability, or retransmission guarantee.*

Point	TCP	UDP
Connection	Connection-oriented	Connectionless
Ordering	Guaranteed	Not guaranteed
Reliability	Retransmits lost data	No retransmission
Speed	Slower (acknowledgements, state)	Faster (light header)
Typical use	Web pages, mail, file transfer	Live voice, video, games, DNS lookups

The exam tests this pair by swapping one property: an option that says UDP guarantees order, or that TCP is connectionless, is always wrong. The anchor is the direction of the trade: TCP buys correctness at the cost of speed, while UDP buys speed at the cost of correctness.

Example 62.2.1. *An item states, “UDP guarantees ordered delivery of datagrams.” This is wrong. Ordering is a TCP guarantee; UDP provides no ordering at all. The wrong option has moved a TCP property onto UDP.*

62.3 ICMP: the network's own messenger

While TCP and UDP carry user data, **ICMP** (Internet Control Message Protocol) carries messages *about* the network. Routers and hosts use it to report errors — destination unreachable, time exceeded, parameter problems — and diagnostic tools such as **ping** (echo request and reply) and **traceroute** are built on ICMP messages.

Definition (ICMP). *The control and diagnostic protocol of the Internet layer; reports errors and supports ping and traceroute, and never carries application data.*

Three facts together close every ICMP trap (PYQ-18, PYQ-19, TRAP-19). First, ICMP transfers no application data: it never moves a file, a mail message, or a web page. Second, it uses no ports: ports belong to TCP and UDP, and an option that assigns ICMP a port number is wrong. Third, it is neither a routing protocol like BGP nor a mail protocol like SMTP; it only reports on conditions the other protocols encounter.

62.4 BGP: routing between autonomous systems

The Internet is not one network but a federation of large independently managed networks called **autonomous systems** (an ISP, a university backbone, a government network). **BGP** (Border Gateway Protocol) is the protocol that routes *between* them. It exchanges reachability information across autonomous-system borders and selects the paths that traffic takes over the Internet backbone.

Definition (BGP). *The inter-domain routing protocol that exchanges reachability between autonomous systems on the Internet backbone.*

The tested contrast is BGP against the *interior* protocols **OSPF** (Open Shortest Path First) and **RIP** (Routing Information Protocol), which route *inside* a single domain (PYQ-19, TRAP-20). Any option that places BGP inside one small local network has swapped the two scopes. The one-word shortcut is “inter-domain”: whenever the stem says inter-domain or between autonomous systems, the answer is BGP.

62.5 IPv6: address space, header, and no broadcast

IPv4 uses 32-bit addresses (about 4.3 billion), which ran out under the growth of the Internet. **IPv6** replaces them with **128-bit** addresses, written as eight groups of hexadecimal digits, giving about 3.4×10^{38} addresses. This vast space is the standard reason given for IPv6 removing the need for address-sharing workarounds such as **NAT** (Network Address Translation), which lets many private addresses share one public IPv4 address (PYQ-19).

Definition (IPv6 address). *A 128-bit Internet-layer address, written as eight hexadecimal groups, with a fixed 40-byte header.*

Two further IPv6 facts are tested. The fixed IPv6 header is **40 bytes** long (PYQ-18). And IPv6 has **no broadcast**: IPv4’s three delivery styles (unicast to one receiver, multicast to a group, broadcast to everyone) become unicast, multicast, and anycast (nearest of several receivers) in IPv6, with a multicast to the all-nodes group replacing the broadcast. An option mentioning an “IPv6 broadcast address” is therefore self-defeating. The 256-bit figure that appears as a distractor is always wrong (TRAP-18).

Example 62.5.1. *An item offers “IPv6 uses 256-bit addresses” as one option. This is wrong. IPv6 uses 128-bit addresses; 256 is double the correct size and matches no Internet-layer standard. The option confuses IPv6 with an unrelated larger number.*

62.6 HTTP/3 and QUIC

Web traffic travelled over TCP for decades: HTTP/1.1 and HTTP/2 both run over TCP. **HTTP/3** breaks that pattern. It runs over **QUIC** (Quick UDP Internet Connections), which itself runs over **UDP**, not over TCP. QUIC rebuilds the needed reliability, congestion control, and security on top of UDP, gaining faster connection setup and independent streams without head-of-line blocking across streams.

Definition (QUIC). *A UDP-based transport protocol carrying HTTP/3, providing its own reliability and security instead of using TCP.*

The trap writes itself: any option claiming HTTP/3 relies on TCP has the protocol stack backwards (PYQ-19, TRAP-21). The correct chain is HTTP/3 over QUIC over UDP.

62.7 DNS and the .org / PIR / ICANN chain

DNS (Domain Name System) is the Internet's directory: it translates human-readable domain names such as `example.org` into numeric IP addresses that routers can use. Resolution runs hierarchically from root servers down through top-level-domain servers to the domain's own authoritative servers. DNS is a lookup service, not a routing protocol and not a mail-transfer protocol, and confusing it with either is a standard wrong answer.

The governance half of the topic names three actors. **.org** is a generic top-level domain originally meant for non-profit organisations. Its registry is operated by **PIR** (Public Interest Registry). **ICANN** (Internet Corporation for Assigned Names and Numbers) coordinates domain names and IP addresses at the global level. The exam distinction is role-based: PIR *runs* the .org registry, while ICANN *coordinates* the naming system as a whole.

62.8 ARPANET and packet switching

ARPANET (Advanced Research Projects Agency Network), live in **1969**, is the ancestor of the Internet. Its contribution was proving that **packet switching** works across long distances: splitting data into addressed packets that travel independently and are reassembled at the destination, with the network routing around failed links. Later ARPANET work produced the TCP/IP protocol family.

Definition (Packet switching). *Splitting data into addressed packets that may take different routes and are reassembled at the destination.*

The tested contrast is packet switching against **circuit switching**, the older telephone model that holds one dedicated path for the whole call. The Internet uses packet switching; the traditional telephone call used circuit switching. Options that assign a dedicated reserved circuit to the Internet have the two models swapped.

62.9 The router: joining networks by reading IP addresses

A **router** connects different networks and forwards packets between them. It does this by reading each packet's destination **IP address** and choosing the next hop from its **routing table**. In assertion–reason form: the assertion (a router joins networks) and the reason (it reads the destination IP and consults its table) are both true, and the reason correctly explains the assertion, because address-reading is *how* the router forwards between networks.

The near-neighbours are the **switch**, which forwards *inside* one network using MAC addresses, and the **hub**, which merely repeats everything it receives. An option that says a router forwards within one network using MAC addresses describes a switch, not a router.

62.10 IaaS: the provider manages the hardware

Cloud service models split responsibility between provider and customer. In **IaaS** (Infrastructure as a Service), the **provider** manages the physical infrastructure: servers, storage hardware, **networking hardware**, and virtualisation. The **customer** manages everything above that layer: the operating system, runtime, applications, and data. The exam anchor is the provider's side: in IaaS the provider manages the networking hardware (PYQ-20).

Definition (IaaS responsibility split). *The provider manages infrastructure (networking and storage hardware, virtualisation); the customer manages the operating system, runtime, applications, and data.*

Moving down the ladder, the provider manages more. In **PaaS** (Platform as a Service) the provider additionally manages the operating system and runtime, leaving the customer its applications and data. In **SaaS** (Software as a Service) the provider manages everything up to the finished application, and the customer only manages its own data and settings. The trap claims the IaaS provider manages the customer’s OS or data; it never does (TRAP-22).

Example 62.10.1. *An item states, “In IaaS the provider manages the customer’s operating system and data.” This is wrong. Those belong to the customer; the provider stops at the infrastructure layer (networking and storage hardware plus virtualisation). The option describes SaaS, not IaaS.*

62.11 Exam retrieval and common confusions

Seven distinctions prevent most errors on this topic.

1. TCP guarantees **order and reliability** but no timing; UDP guarantees **neither** (PYQ-18).
2. ICMP carries **control messages only**: no application data, no ports (TRAP-19).
3. BGP is **inter-domain** (between autonomous systems); OSPF and RIP are intra-domain (TRAP-20).
4. IPv6 is **128-bit** with a **40-byte** header; 256-bit is always wrong (PYQ-18, TRAP-18).
5. IPv6 has **no broadcast**; the replacement is **multicast** to the all-nodes group (PYQ-19).
6. HTTP/3 uses **QUIC over UDP**, never TCP (TRAP-21).
7. In IaaS the **provider** manages networking hardware; the **customer** manages the OS, applications, and data (TRAP-22).

Example 62.11.1. *An item asks which protocol routes between autonomous systems on the Internet backbone. Trace the scope word: “between autonomous systems” is the definition of inter-domain routing, and the inter-domain protocol is BGP. OSPF and RIP are interior protocols, and ICMP is not a routing protocol at all, so all three are eliminated by scope.*

Exercises

1. State the two guarantees TCP provides and the one it does not.
2. Distinguish TCP from UDP on connection, ordering, reliability, and speed.
3. What does ICMP carry, and what two things does it never do?
4. Distinguish BGP from OSPF/RIP by scope.
5. State the IPv6 address size and header size, and explain what replaces the broadcast.
6. Explain the HTTP/3 protocol stack in one chain of three names.
7. Distinguish the roles of PIR and ICANN for the .org domain.
8. State the IaaS responsibility split in two sentences.

Solutions

1. TCP guarantees ordered delivery and reliable delivery (lost segments are retransmitted). It provides no timing guarantee.
2. TCP is connection-oriented, ordered, and reliable but slower; UDP is connectionless, unordered, and unreliable (no retransmission) but faster.
3. ICMP carries control and diagnostic messages (errors, ping, traceroute). It never transfers application data and never uses ports.
4. BGP routes between autonomous systems (inter-domain, Internet backbone); OSPF and RIP route inside a single domain.
5. IPv6 uses 128-bit addresses with a fixed 40-byte header. It has no broadcast; a multicast to the all-nodes group replaces it.
6. HTTP/3 runs over QUIC, which runs over UDP: HTTP/3 over QUIC over UDP.
7. PIR operates the .org registry; ICANN coordinates domain names and IP addresses globally.
8. The provider manages the infrastructure (networking and storage hardware, virtualisation); the customer manages the operating system, runtime, applications, and data.